
Data Analytics
Academic Year 2025-26
Course Project N.03: Clinical Records
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For this assignment you will work individually to carry out simple tasks of data analysis given a specific dataset. The goal of this assignment is to use Python and complementary libraries on a given dataset in order to *explore* and *analyze* the given data and *draw conclusions*.

Description

The data lists various attributes of people diagnosed with diabetes from the year 1999 to 2008 in 130 US hospitals. The length of stay of each of these people at a hospital is at least 1 day and at most 14 days. The dataset contains attributes such as patient number, race, gender, age, admission type, number of lab tests performed, diagnosis, medications, etc. This dataset contains a set of different features of the people diagnosed with diabetes. Your goal is to cluster these patients based on their various attributes. For example, what are different categories of patients based on the type of their prescribed medication? Your tasks are to:

- Explore and describe the data (*i.e.*, standard descriptive statistics, visualize the variables with different graphs, draw distributions and histograms of variables, are there outliers? Any interesting observation? Any correlations? Etc.)
- Pre-process the data (*i.e.*, handle and fill unknowns if there are any, etc.)
- Use at least one different clustering algorithm and analyze the results. What is the most optimal number of clusters?
- Evaluate the performance of the model.

Submission procedure and evaluation

You should produce a report of your work and its evaluation along with the source code. It will be a concise explanation of how you tackled the different tasks, the reasons of your choices, successive conclusions, graphs you produced, results of the decisions and their accuracy, *etc.*

Use Jupyter Notebook to produce results of the commands in a single .ipynb file. For more information check: <https://jupyter.org/documentation>

The report (max 5 pages) and the code of the project need to be submitted via iCorsi.

Please, upload all the required items in a single file and name it following the structure: **no_Project.[zip|tar.gz|7z]**. For instance, 05_projectname.tar.gz